

Software Requirements Specification (SRS)

TasteBud Recipes : Recipe Sharing Platform

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document outlines the functional and non-functional requirements for TasteBud Recipes, a web-based platform that allows users to create, share, discover, and rate recipes in a simple and modern way. It helps home cooks keep their favorite recipes organized and easy to find, while also connecting with other cooking enthusiasts. The system offers easy recipe management, powerful search features, and a friendly rating system to make finding and sharing recipes enjoyable and accessible for everyone.

1.2 Scope

TasteBud Recipes is a web application that will:

- Allow users to create and share recipes (with ingredients, steps, and cooking details)
- Enable food lovers to browse and search recipes (by name, ingredients, or category)
- Let users rate recipes (1-5 stars to help others find the best ones)
- Provide user profiles (to manage personal recipes)
- Give admins basic control (to remove inappropriate content)

Coming later:

- Recipe photos
- Comments and reviews
- Social sharing

1.3 Definitions and Acronyms

1.3.1 Definitions:

- **Recipe:** A structured cooking guide including ingredients, steps, and preparation details.
- **User:** A registered member who can create, browse, or rate recipes.
- **Admin:** A privileged user who manages content and user accounts.
- **Rating:** A 1-5 star score given by users to recipes.
- **Category:** Recipe classification (e.g., Breakfast, Dessert).

1.3.2 Acronyms

- **CRUD:** Create, Read, Update, Delete
 - **MVC:** Model-View-Controller (ASP.NET architecture)
 - **UI/UX:** User Interface / User Experience
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2. Overall Description

2.1 Product Perspective

TasteBud Recipes is an innovative web-based platform designed to revolutionize how cooking enthusiasts share and discover recipes. Built as a standalone system using modern web technologies, it serves as a digital cookbook that combines:

- **Recipe Management:** Structured format for ingredients and instructions
- **Social Features:** Rating system for community validation
- **Discovery Tools:** Advanced search capabilities

The system integrates:

- **Frontend:** ASP.NET MVC with Bootstrap 5 for responsive design
- **Backend:** ASP.NET Web API for future mobile expansion
- **Database:** SQL Server with Entity Framework Core
- **Authentication:** Secure ASP.NET Identity system

2.2 User Classes and Characteristics

1. Guest Users

- Browse and search all public recipes
- View recipe details and ratings
- No contribution capabilities

2. Registered Users

- Create, edit, and manage personal recipes
- Rate other users' recipes (1-5 stars)
- Build personal profile and recipe collection
- Use advanced search filters

3. Administrators

- Moderate all content (recipes and users)
- Access comprehensive dashboard
- Handle user reports and disputes
- Configure system settings

2.3 Operating Environment

Technical Stack:

- **Application Server:** Windows Server with IIS or cross-platform Kestrel
- **Database Server:** Microsoft SQL Server 2019+
- **Development Framework:** .NET 6+

Frontend Technologies:

- HTML5, CSS3, JavaScript (ES6+)
- Bootstrap 5 for responsive layouts
- jQuery for AJAX functionality

Supported Environments:

- **Browsers:** Latest versions of Chrome, Firefox, Edge, and Safari
- **Devices:** Fully responsive for desktop, tablet, and mobile

3. Functional Requirements

3.1 User Management

1. User Registration

- Allow sign-up via email and password (minimum 8 characters).

2. User Login/Logout

- Secure authentication with session management.

3. Profile Management

- Users can view and edit their display name and basic profile details.

4. Password Reset

- Email-based reset link for forgotten passwords.

3.2 Recipe Management

5. Create Recipe

- Users can submit recipes with:
 - Title and description
 - Ingredients (with quantities)
 - Step-by-step instructions
 - Prep time, cook time, and servings
 - Difficulty level (Easy/Medium/Hard)
 - Category (e.g., Breakfast, Dessert)

6. Edit Recipe

- Authors can modify their own recipes.

7. Delete Recipe

- Authors can remove their recipes (with confirmation prompt).

8. View Recipe

- Publicly accessible detailed view with all metadata.

3.3 Search & Discovery

9. Search by Keyword

- Fuzzy search functionality for recipe titles and descriptions.

10. Filter by Ingredient

- Users can search for recipes containing specific ingredients.

11. Filter by Category

- Dropdown to filter recipes by meal type (e.g., Lunch, Snacks).

12. Sort Options

- Sort recipes by:
 - Newest first
 - Highest-rated
 - Shortest cooking time

3.4 Rating System

13. Rate Recipes

- Logged-in users can assign 1–5 star ratings (whole numbers only).

14. Display Average Rating

- Show calculated average (rounded to 1 decimal) on recipe pages.

15. Prevent Duplicate Ratings

- One rating per user per recipe.

3.5 Administrative Functions

16. Content Moderation

- Admins can delete recipes or users (no pre-approval in v1).

17. View All Users

- Admin dashboard with user list and activity metrics.

18. System Configuration

- Basic settings (e.g., default categories).

4. Non-Functional Requirements

4.1 Performance Requirements

Response Time: All pages load within 3 seconds

Concurrent Users: Supports 500+ active users simultaneously

Search Speed: Recipe searches return results in ≤ 2 seconds

Mobile Friendly: Works smoothly on phones and tablets

4.2 Security Requirements

Passwords: Stored securely with hashing + salting

Sessions: Auto-logout after 30 minutes inactive

Input Safety: All user inputs filtered for malicious code

Data Protection: HTTPS encryption for all communications

4.3 Reliability Requirements

Uptime: 99% availability (max 7hrs downtime/month)

Backups: Automatic daily database backups

Recovery: Critical failures fixed within 4 hours

4.4 Usability Requirements

Easy Learning: New users can post recipes in <5 minutes

Clear Errors: Simple messages explain what went wrong

Consistent Design: Same buttons/actions across all pages

4.5 Compatibility Requirements

Browsers: Works on Chrome, Firefox, Edge, Safari

Devices: Adapts to phone/tablet/desktop screens

4.6 Maintenance Requirements

Clean Code: Follows standard .NET coding practices

Testing: Core features have automated tests

5. Use Cases

5.1 User Registration

Actor: Guest

Description: A new user creates an account to access recipe-sharing features.

Steps:

1. Guest clicks "Sign Up"
2. Enters email, password, and display name
3. Receives confirmation email (optional in v1)
4. Account is created automatically

5.2 User Login

Actor: Registered User

Description: An existing user accesses their account.

Steps:

1. User enters email and password
2. System verifies credentials
3. Session starts with personalized homepage

5.3 Create Recipe

Actor: Registered User

Description: User shares a new recipe with the community.

Steps:

1. User clicks "Add Recipe"
 2. Fills in:
 - Title (e.g., "Grandma's Apple Pie")
 - Description (story/notes)
 - Ingredients + quantities
 - Step-by-step instructions
 - Prep/cook time, servings, difficulty
 3. Submits → Recipe appears publicly
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5.4 Search Recipes

Actor: Any User

Description: Finds recipes matching specific criteria.

Steps:

1. User types search term (e.g., "chicken")
 2. Filters by:
 - Ingredients (e.g., "tomatoes")
 - Category (e.g., "Dinner")
 - Difficulty (e.g., "Easy")
 3. Views matching results
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5.5 Rate Recipe

Actor: Registered User

Description: Rates a recipe after trying it.

Steps:

1. User views recipe details
 2. Selects 1-5 stars
 3. Submits → Rating updates average score
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5.6 Manage Recipes (Edit/Delete)

Actor: Recipe Owner

Description: Updates or removes their recipe.

Steps for Edit:

1. Owner clicks "Edit" on their recipe
2. Makes changes
3. Saves → Updates appear immediately

Steps for Delete:

1. Owner clicks "Delete"
2. Confirms action
3. Recipe is permanently removed